1. To find the magnetic field within the magnetic material (between the tubes) always use Ampere's law in $\vec{H}$ (auxiliary field), $\oint \vec{H}\cdot d\vec{l} = (I_f)_{enc}$ —— (i)

In the problem as the currents are longitudinal, the fields $\vec{B}$ as well as $\vec{H}$ must be circumferential. Applying (i) to <u>any</u> Amperian loop of radius 's' gives $\boxed{\vec{H} = \frac{I}{2\pi s}\hat{\varphi}}$ —— (ii)

For $a < s < b$, magnetization would be given by $\vec{M} = \chi_m \vec{H}$. [Note that this is the only magnetic region and hence M is non-zero only in this region]

$$\boxed{\vec{M} = \frac{I\chi_m}{2\pi s}\hat{\varphi}} \quad \text{—— (iii)}$$

The total field $\vec{B}$ is given by $\vec{B} = \mu_0(\vec{H}+\vec{M})$ [derived in class]

$$\vec{B} = \mu_0(\vec{H} + \chi_m\vec{H}) = \mu_0(1+\chi_m)\vec{H}$$

$$\boxed{\vec{B} = \mu_0(1+\chi_m)\frac{I}{2\pi s}\hat{\varphi}} \quad \text{—— (iv)}$$

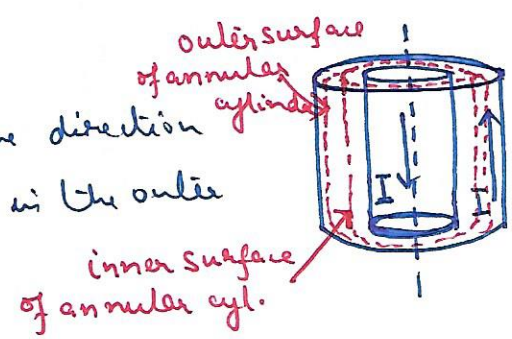
This is the total field in the region between the tubes.

Now we will verify (iv) using field due to both the free and bound currents.

First, let's calculate the bound currents $\vec{J}_b = \vec{\nabla}\times\vec{M}$. From (iii) $M = M_\varphi(s)$. Using expression of curl in cylindrical co-ordinates, $\vec{\nabla}\times\vec{M} = 0$, hence $\vec{J}_b = 0$

$\vec{K}_b = \vec{M}\times\hat{n}$ is the surface current.

In the region between the tubes, the direction of surface currents in the inner surface and in the outer surface are opposite.

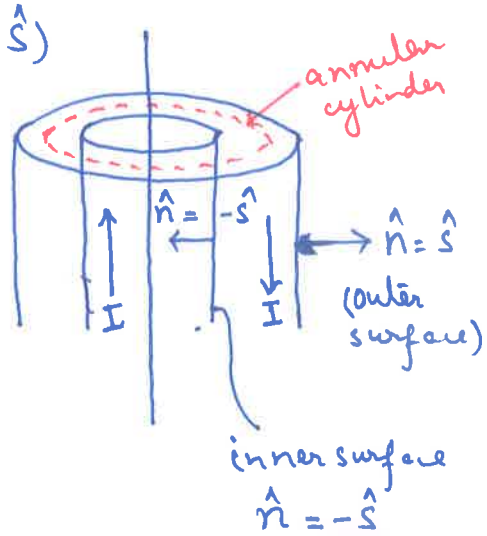


For inner surface of annular cylinder

$$\vec{K}_b = \vec{M} \times \hat{n} = M(\hat{\varphi}) \times (-\hat{s})$$

$$= M\,\hat{z}$$

$$= \left.\frac{I\chi_m}{2\pi s}\right|_{s=a} \hat{z}$$



$$\vec{K}_b = \frac{I\chi_m}{2\pi a}(\hat{z})$$

Similarly for outer surface of annular cylinder

$$\vec{K}_b = \vec{M} \times \hat{n} = M(\hat{\varphi}) \times (\hat{s}) \quad [\hat{n} = \hat{s}]$$

$$= -M\,\hat{z}$$

$$= \left.\frac{I\chi_m}{2\pi s}\right|_{s=b} (-\hat{z}) = -\frac{I\chi_m}{2\pi b}(\hat{z})$$

To verify $\vec{B}$ in the <u>annular region</u> in (iv), use

$$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc} \qquad I_{enc} = I_f + \oint J_b\,da + \int K_b\,dl$$

$$= I + \frac{\chi_m I}{2\pi a} \times 2\pi a$$

$$B \cdot 2\pi s = \mu_0 I(1+\chi_m)$$

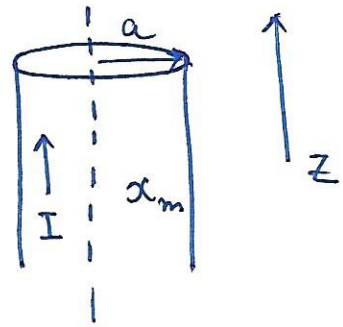
$$= I(1+\chi_m)$$

$$B = \frac{\mu_0 I(1+\chi_m)}{2\pi s}$$

— same as in (iv)

2. Again, this is a problem of magnetic material - so use

$$\oint \vec{H}\cdot d\vec{l} = (I_f)_{enc}$$



Clearly H is circumferential - consider an Amperian loop of radius 's'

<u>Inside cylinder $s < a$</u>

$$H \cdot 2\pi s = I \times \frac{\pi s^2}{\pi a^2}$$

$$\vec{H} = \frac{I s}{2\pi a^2}\,\hat{\varphi} \quad \text{———} \quad s < a$$

<u>Outside cylinder $s > a$</u>

$$H \cdot 2\pi s = I$$

$$\boxed{\vec{H} = \frac{I}{2\pi s}\,\hat{\varphi}} \quad \text{———} \quad s > a$$

Magnetization, $\vec{M} = \chi_m \vec{H}$

$$\boxed{\vec{M} = \frac{\chi_m I s}{2\pi a^2}\,\hat{\varphi}}$$

The total field $\vec{B} = \mu_0 (\vec{H} + \vec{M})$

$$\vec{B} = \left(\frac{\mu_0 (1+\chi_m) I s}{2\pi a^2}\right)\hat{\varphi} \qquad s < a$$

$$= \frac{\mu_0 I}{2\pi s}\,\hat{\varphi} \qquad s > a$$

<u>Bound currents</u>

$$\vec{J}_b = \vec{\nabla}\times\vec{M} = \vec{\nabla}\times \chi_m \vec{H} = \chi_m (\vec{\nabla}\times\vec{H}) = \chi_m \vec{J}_f \quad [\vec{J}_f = \text{free current density}]$$

Since in the problem, current is uniformly distributed

$$\cancel{I_b = \int J_b\, da = \chi_m \int J_f\, da = \chi_m}$$

$$\vec{J}_b = \chi_m \vec{J}_f = \chi_m \frac{I}{\pi a^2}(\hat{z})$$

Surface current, $\vec{K}_b = \vec{M}\times\hat{n} = \chi_m(\vec{H}\times\hat{n}) = \left.\frac{\chi_m I s}{2\pi a^2}\right|_{s=a} (\hat{\varphi}\times\hat{s})$

$$\vec{K}_b = \frac{\chi_m I}{2\pi a}(-\hat{z})$$

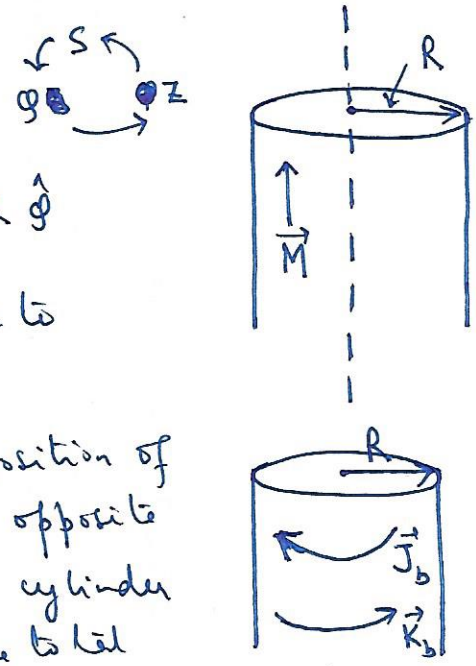
Thus, we have $\vec{J}_b = \frac{\chi_m I}{\pi a^2}\hat{z}$ and $\vec{K}_b = \frac{\chi_m I}{2\pi a}(-\hat{z})$. Thus the volume current runs parallel to free current I but surface current runs antiparallel to I – this happens when $\chi_m > 0$. (paramagnetic material). The reverse happens for $\chi_m < 0$ (diamagnetic materials)

Net bound current $= \int J_b \, da + \int K_b \, dl$

$$= \frac{\chi_m I}{\pi a^2} \times \pi a^2 - \frac{\chi_m I}{2\pi a} \times 2\pi a = 0$$

This is quite obvious, net bound currents are always zero.

3. $\vec{M} = ks\hat{z}$



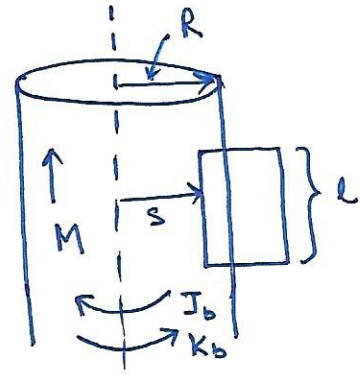
(a) $\vec{J}_b = \vec{\nabla} \times \vec{M} = -k\hat{\varphi}$

$\vec{K}_b = \vec{M} \times \hat{n} = ks(\hat{z} \times \hat{s})\big|_{s=R} = kR\hat{\varphi}$

[For curls in <u>cylindrical co-ord</u> – refer to Griffiths front cover page]

Now this case is equivalent to superposition of two solenoids carrying currents in opposite directions. So the <u>field outside</u> the cylinder must be zero. This is because the total bound current due to the two bound current densities must be zero.

Field inside cylinder



Since $\vec{B}$ must be longitudinal ($\vec{M}$ is longitudinal) - consider an rectangular Amperian loop whose inner edge is at a distance 's' from the axis.

Apply Ampere's law $\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$ to find $\vec{B}$. Here I_{enc} refers to the total current due to bound currents flowing along the closed rectangular loop.

$$I_{enc} = [J_b (R-s) \times l + K_b \times l]$$

$$\therefore \quad B_{in} \times l = [-k(R-s)\, l + kRl] \times \mu_0 = ksl \times \mu_0 \quad [B_{out} = 0]$$

$$\vec{B}_{in} = ks\,\hat{z} \times \mu_0$$

$$\boxed{\vec{B}_{in} = \mu_0 k s \hat{z}}$$

(b) Ampere's law in $\vec{H}$ gives $\oint \vec{H}\cdot d\vec{l} = (I_f)_{enc}$. As $\vec{H}$ points along z-axis, same Amperian loop (as above) gives

$$H \times l = 0 \quad [\text{no free current}, I_f = 0 \text{ in problem}]$$

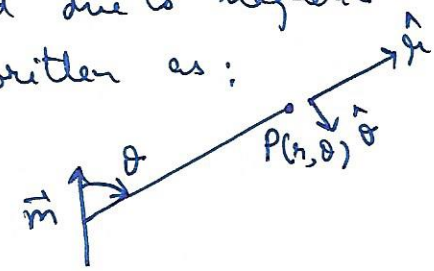
$$\vec{H} = \vec{0} \Rightarrow \vec{B} = \mu_0(\vec{H} + \vec{M}) = \mu_0 \vec{M}$$

<u>Inside:</u> $\vec{B} = \mu_0 \vec{M} = \mu_0 k s (\hat{z})$ [as obtained before]

<u>Outside:</u> $\vec{B} = \mu_0 \vec{M} = \vec{0}$ [$\vec{M} = 0$ outside] "

4. To prove the interaction energy between two magnetic

(a) dipoles, recall that ~~a~~ ~~mag~~ magnetic field due to magnetic dipole aligned along z-axis can be written as:



$$\vec{B}(r,\theta) = \frac{\mu_0}{4\pi}\frac{1}{r^3}\left[2m\cos\theta\,(\hat{r}) + m\sin\theta\,(\hat{\theta})\right] \quad \text{— (vii)}$$

(proved in class)

The term in square brackets can be written in a co-ordinate free form as $\left[3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}\right]$. This follows from expressing $\vec{m}$ as

$$\vec{m} = m_r\hat{r} + m_\theta\hat{\theta} = (\vec{m}\cdot\hat{r})\hat{r} + (m\cdot\hat{\theta})\hat{\theta}$$

$$= (m\cos\theta)\hat{r} - (m\sin\theta)\hat{\theta}$$

[refer fig. above]

Now $3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}$

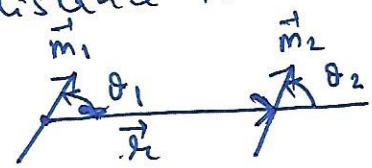
$$= 3(m\cos\theta)\hat{r} - (m\cos\theta)\hat{r} + (m\sin\theta)\hat{\theta}$$

$$= (2m\cos\theta)\hat{r} + (m\sin\theta)\hat{\theta} \quad \text{— (viii)}$$

Using (vii), (viii)

$$\boxed{\vec{B}(r,\theta) = \frac{\mu_0}{4\pi}\frac{1}{r^3}\left[3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}\right]} \quad \text{— (ix)}$$

Now if two dipoles are separated by distance '$\vec{r}$' as given in the question



$\vec{B}_1$ (due to $\vec{m}_1$) at position of $\vec{m}_2$ will be given by (ix) and replacing $\vec{m} \to \vec{m}_1$. The interaction energy (potential energy) is then given by $U = -\vec{m}_2\cdot\vec{B}_1$ [$\because U = -\vec{m}\cdot\vec{B}$ done in class]

This gives, $U = \frac{\mu_0}{4\pi}\frac{1}{r^3}\left[\vec{m}_1\cdot\vec{m}_2 - 3(\vec{m}_1\cdot\hat{r})(\vec{m}_2\cdot\hat{r})\right]$

(b) Let's express the interaction energy in terms of angles θ_1 and θ_2.

$$U = \frac{\mu_0}{4\pi}\frac{1}{r^3} m_1 m_2 \left[\cos(\theta_2-\theta_1) - 3\cos\theta_1\cos\theta_2\right] \quad \left[\text{assume } \theta_2 > \theta_1 \text{ from (ix)}\right]$$

$$= \frac{\mu_0}{4\pi}\frac{1}{r^3} m_1 m_2 \left[\sin\theta_2 \sin\theta_1 - 2\cos\theta_1\cos\theta_2\right] \quad (X)$$

$$\left[\cos(\theta_2-\theta_1) = \cos\theta_2\cos\theta_1 + \sin\theta_2\sin\theta_1\right]$$

Since dipole is free to rotate (given in question) — stable configuration is given by $\frac{\partial U}{\partial \theta_1} = \frac{\partial U}{\partial \theta_2} = 0$ [$\vec{r}$ is fixed]

$$\frac{\partial U}{\partial \theta_1} = 0 \Rightarrow \sin\theta_2\cos\theta_1 + 2\sin\theta_1\cos\theta_2 = 0$$

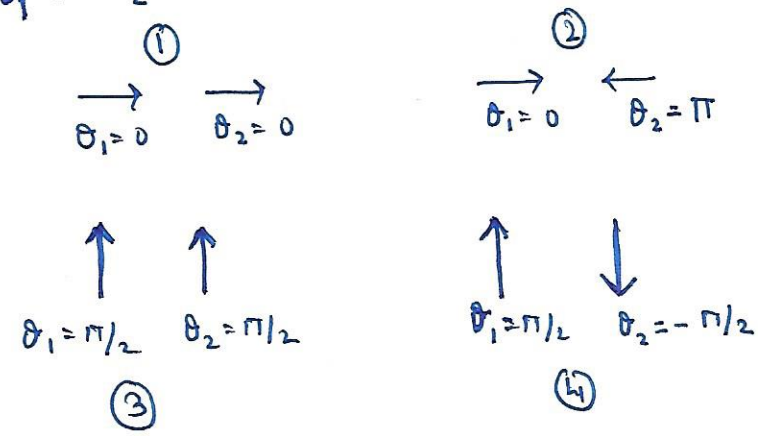
$$\Rightarrow \sin\theta_2\cos\theta_1 = -2\sin\theta_1\cos\theta_2 \quad \text{(xi a)}$$

$$\frac{\partial U}{\partial \theta_2} = 0 \Rightarrow 2\sin\theta_1\cos\theta_2 = -4\sin\theta_2\cos\theta_1 \quad \text{(xi b)}$$

(xia), (xib) gives $\sin\theta_1\cos\theta_2 = 0 = \sin\theta_2\cos\theta_1$

$\sin\theta_1 = \sin\theta_2 = 0$

$\cos\theta_1 = \cos\theta_2 = 0$



Quantitatively, using eqn. (X) energy is minimum for configuration 1

$U_{①} = \frac{\mu_0}{4\pi}\frac{-2m_1m_2}{r^3}$. For e.g. for config. ④, $U_{④} = -\frac{\mu_0}{4\pi}\frac{m_1m_2}{r^3}$

Qualitatively, this means that for configurations ②, ③, ④, $\vec{m}_2$ is not aligned along $\vec{B}_1$ (field due to $\vec{m}_1$)

$$\vec{B}_1 = \frac{\mu_0}{4\pi}\left(\frac{1}{r^3}\right)\left[3\underbrace{(\vec{m}_1\cdot\hat{r})\,\hat{r}}_{\text{field along } \hat{r}} - \vec{m}_1\right]$$

$\vec{m}_2$ is aligned along $\vec{B}_1$ only for config. ① – parallel configuration. So we see that due to the interaction energy between two magnetic dipoles – there is a tendency for parallel alignment of two dipoles.

In case of ferromagnetic materials, such parallel alignment of dipoles occur spontaneously (without external field). In actual, quantum mechanical exchange forces are responsible for the phenomenon of ferromagnetism.

For a classical demonstration of ferromagnetism, imagine you had a large collection of compass needles, mounted on pins at regular intervals along a straight line. They would all line up parallel → → → → → along the line joining them.